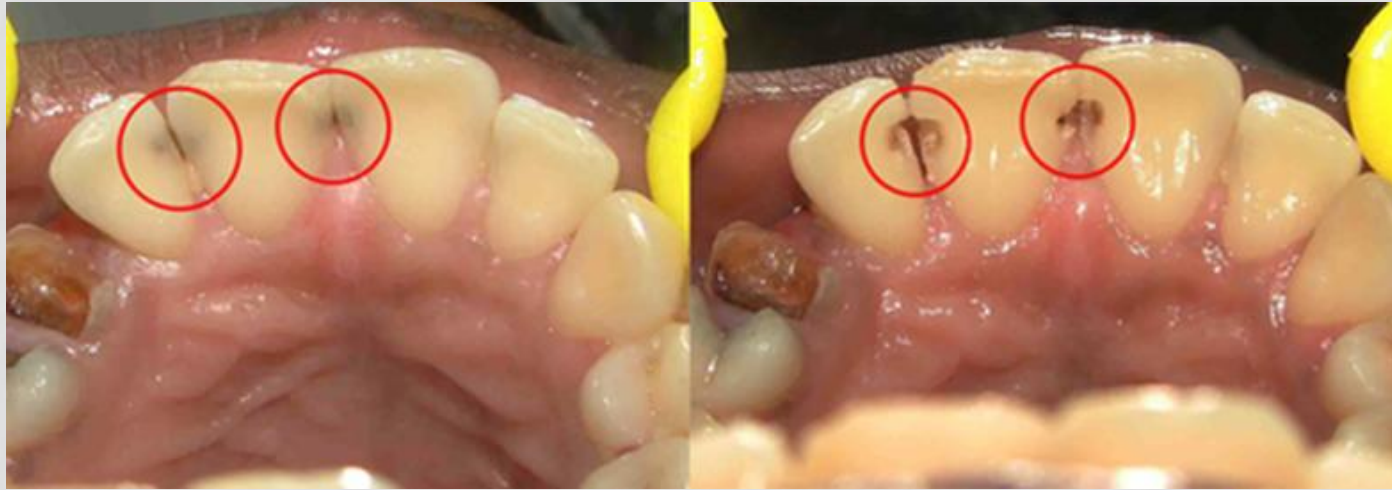
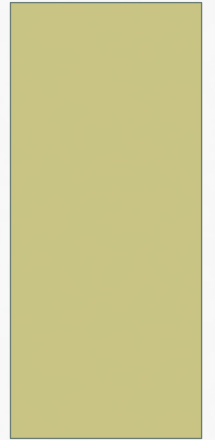




CLASS III CAVITY PREPARATION



LEARNING OUTCOMES

- List the indications for lingual approach and facial approach in a class III cavity preparation
- Explain in detail about the steps of class III cavity preparation

INDICATIONS FOR COMPOSITE

- Almost all Class III and IV restorations are appropriately restored with composite.
- In all of these instances, the operating area must be able to be isolated (**Rubber dam**) adequately to attain an effective bond.

Sturdevant's 5th edition art and science of operative dentistry page no- 530

CONTRAINDICATIONS FOR COMPOSITE

Contraindications for use of composite in class III and IV restorations include

(1) An operating area that cannot be adequately isolated with rubber dam.

(2) Restorations that extend onto the root surface.





The extension onto the root surface (no marginal enamel) may be a contraindication, a V-shaped gap (contraction gap) is formed between the root and the composite.

This contraction gap occurs because the force of **polymerization shrinkage** of the composite

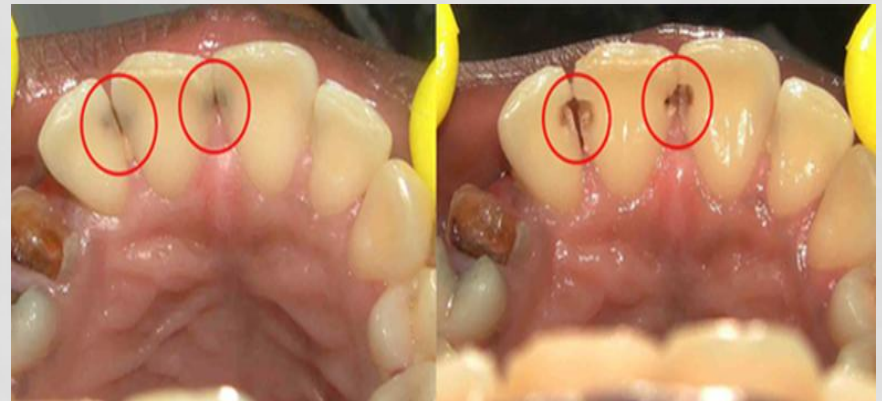
The use of a **resin modified glass-ionomer (RMGIC)** liner on the root surface portion of the preparation may reduce microleakage, gap formation, and recurrent caries.

Tooth preparation for composites should be as **CONSERVATIVE** as possible.

Extent of the cavity preparation is determined by the **SIZE, SHAPE AND LOCATION** of the defect.

CLASS III CAVITY AND COMPOSITE RESTORATION

- Class III preparations seen on the proximal surface of the anterior teeth **without** the involvement of incisal angle.



PREOPERATIVE ASSESSMENT

- Shade selection(First step) is more important.
- Preoperative assessment of occlusion(in class IV).
- Local anaesthesia
- Isolation with rubber dam

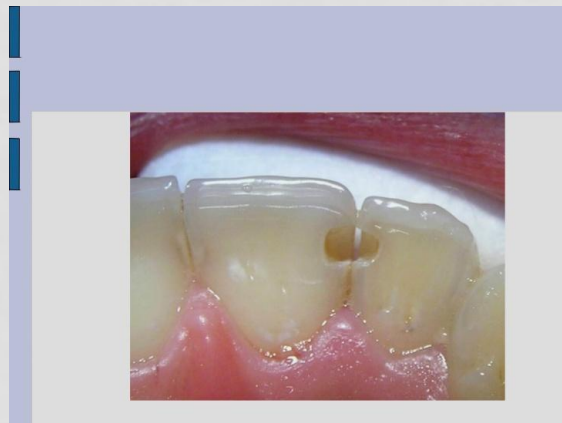
- Composite restorations are primarily retained by the micromechanical bond from acid etching and resin bonding



- When a proximal surface of an anterior tooth is to be restored and there is a choice between facial or lingual entry into the tooth, the **Palatal approach is preferable.**

ADVANTAGES OF RESTORING THE PROXIMAL LESION FROM THE LINGUAL/PALATAL APPROACH

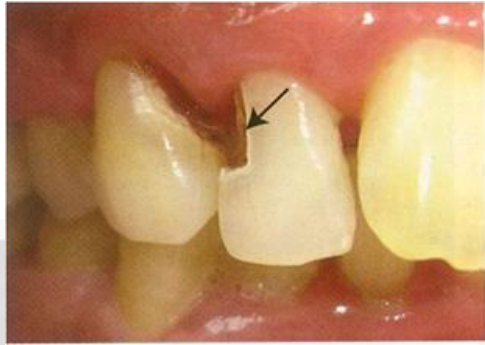
1. The facial enamel is conserved for enhanced esthetics.
2. Color matching of the composite is not as critical.
3. Discoloration or deterioration over the years of the restoration is less visible.





Indications for Facial Approach

1. The carious lesion is positioned facially
2. Teeth is irregularly aligned, making lingual access undesirable.



3. Extensive caries extent into the facial surface.

4. Faulty restoration that was originally placed at the facial.



- When the facial and the lingual surfaces are involved, the approach that provides the best access for instrumentation should be used.



Sturdevant's 5th edition art and science of operative dentistry page no- 531

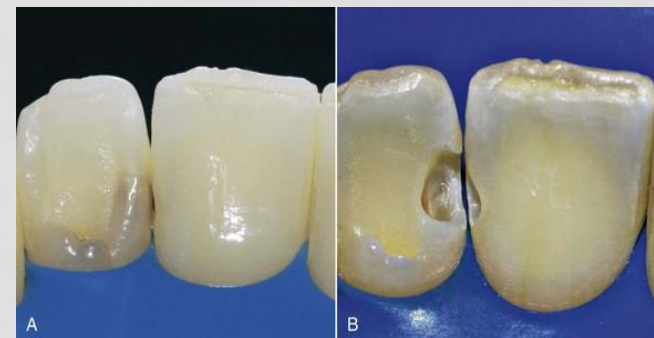
IN CASE OF APPROXIMAL LESIONS (TWO CARIOUS LESIONS JOINING)



- To prepare and restore approximating carious lesions on adjacent teeth at the same appointment.
- Cavity preparation should start with the larger carious teeth first, next the teeth which has less caries should be prepared .
- The reverse order would be followed when the restorative material is inserted.



Restore first smaller cavity followed by large cavity preparation.



DESIGNS OF TOOTH PREPARATION.

There are 3 Designs for class III & IV caries.

1.CONVENTIONAL.

2.BEVELED CONVENTIONAL.

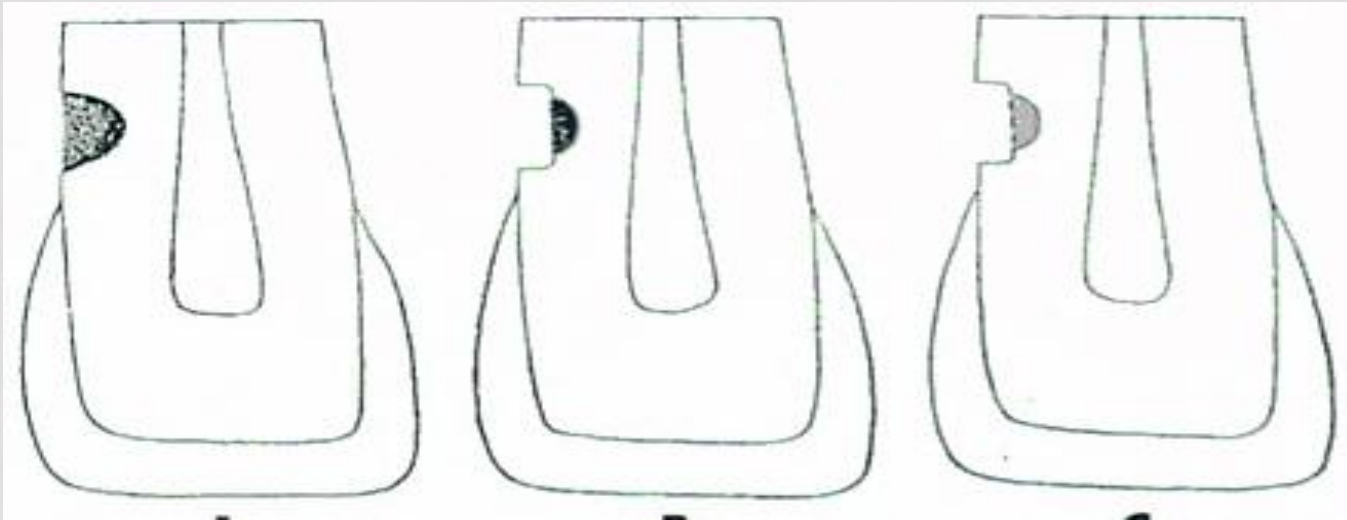
3.MODIFIED.

Sturdevant's 5th edition art and science of operative dentistry page no- 532

1. CONVENTIONAL TYPE.

INDICATIONS.

When caries located on root surfaces (Non-enamel areas).



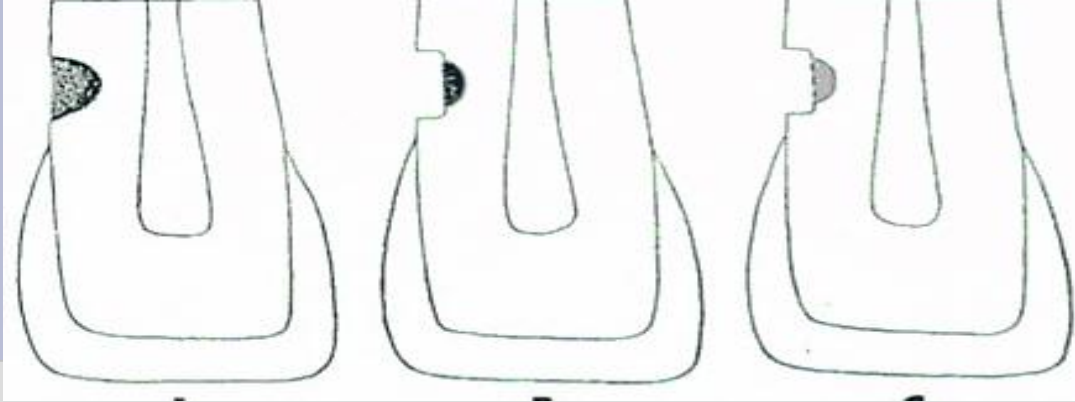
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1. Conventional Class III cavity preparation

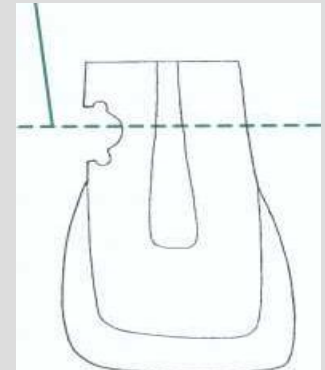
Indicated for restorations involving the root surface

1. Using a No. ½, 1, 2 round bur enter the outline form on the root surface (Bur size varies depends on size of caries)

2. Extend the preparation using straight fissure bur pulpally upto 0.75 mm in depth



3. Use low speed round bur or spoon excavator to remove remaining soft caries.
4. The gingival/cervical and incisal wall is perpendicular to the root surface (box like design)
5. **No bevel is required**
6. A continuous retention groove (secondary retention) is placed when maximum retention required.

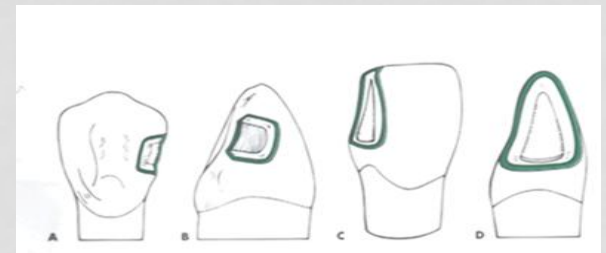


It is placed 0.25 mm in depth parallel to root surface with a $\frac{1}{4}$ round bur.

2. BEVELED CONVENTIONAL

It is similar to Conventional Preparation but with some BEVELED MARGINS.

Mainly indicated when a composite restoration is used to replace an existing old restoration.





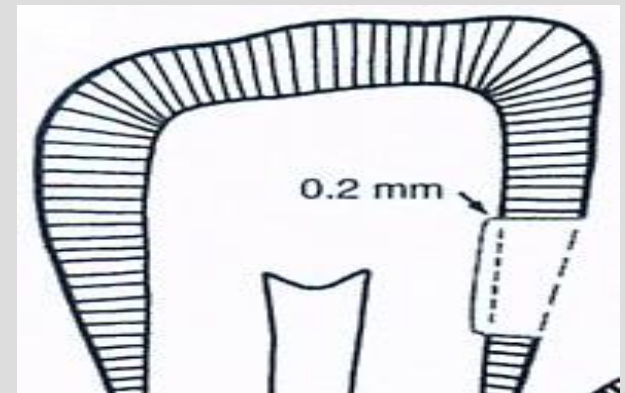
1. Use a round bur No. 1/2, 1, 2 depending on the size of the caries to enlarge the opening sufficiently to allow for caries removal.
2. Extend external walls to sound tooth structure using a straight bur



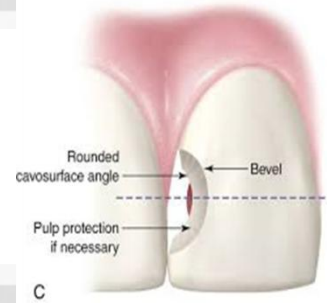
4. Create an axial wall depth of 0.2mm into the dentin/DEJ

(approximately 0.75 – 1.25mm in depth)

5. Axial wall is convex, following the external contour of the tooth.
6. Remove all remaining infected dentin, using a round bur or small spoon excavator.

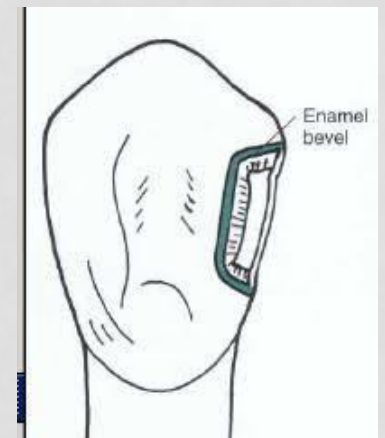


CAVOSURFACE BEVELING



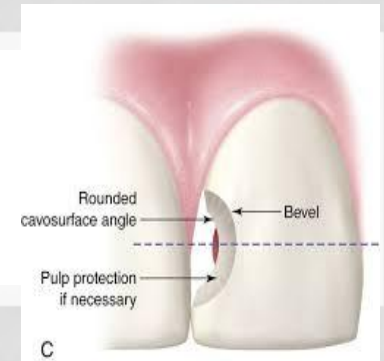
The cavosurface bevel is best prepared with either a flame shaped diamond bur, in an angle approximately 45 degrees to the external tooth surface .

A bevel width of 0.5 mm is considered sufficient.





Beveling Advantages

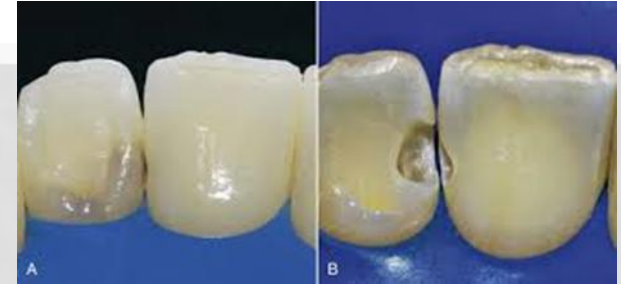
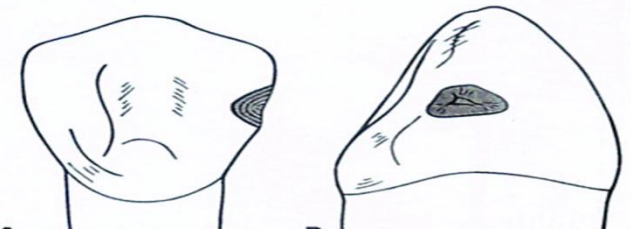


- Increases retention
- Reduces marginal leakage and discoloration of composite restoration.
- Also bevelling helps the restoration to blend more esthetically with the color of the surrounding tooth structure.



CAVOSURFACE BEVELING

- If the preparation extends gingivally onto root structure, no bevel is placed on cementum, as the area is prepared as a conventional preparation.
- In addition, bevels may not be recommended on lingual surface margins that are in areas of centric contact or subjected to heavy masticatory forces because composite has less wear resistance than enamel for withstanding heavy attritional forces.
- Place a bevel only on facial surface enamel, No bevel in lingual enamel or cementum root

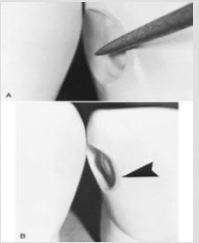


3. MODIFIED PREPARATION.

- Most commonly used
- It is indicated for small and moderate lesions or faults and is designed to be as conservative as possible.
- The preparation design is dictated by the extent of the caries or defect and is prepared from a lingual approach whenever possible
- These are more Conservative preparations and retention is obtained primarily by micromechanical adhesion to the surrounding enamel and dentin.

- Use a 1/2, 1, 2 round bur- initial entry into tooth
- Remove all remaining caries or defect with the straight bur
- Modified preparations are prepared to an initial axial wall depth of 0.2 mm into dentin.

Extent, depth and margins depends on the carious lesion and other defects. These do not have specified wall configurations nor specific pulpal or axial depths.



Resembles scooped out appearance

Bevel done only on facial surface. Use a flame shaped bur resulting in a 45 degrees angle to the external tooth surface.

Ideal Bevel width should be 0.5mm.



Essential reading:

- Sturdevant's 5th edition art and science of operative dentistry
page no- 530 - 540
- Pickard's Manual of Operative Dentistry, 8th edition page no-
156-159

